

Special Problem Defense

MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced
Chimeric Reads in Mitochondrial Illumina Data

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Opening question: which read is artificial?

Opening question: which read appears artificially joined?

Read A



continuous alignment pattern

Read B



possible junction

The task of MitoChime is to automate this decision at read scale.

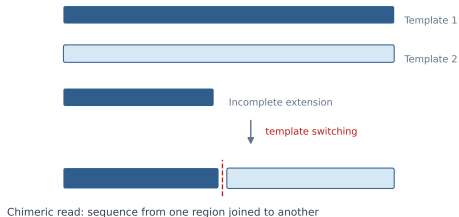
Framing question

Which read shows evidence of being artificially joined?

- ▶ Sequencing datasets contain thousands of reads.
- ▶ Manual inspection is not practical.
- ▶ The goal is automated read-level detection.

Real-world problem

PCR-induced chimera formation



PCR amplification can produce **artificial hybrid reads**.

These reads may be included in sequencing data and mislead mitochondrial genome assembly.

Core issue

Some reads appear valid but do not represent the original mitochondrial sequence.

Why the problem matters

Assembly

Fragmentation

Contigs may break or branch

Analysis

False signals

Incorrect junctions may appear

Interpretation

Lower reliability

Downstream conclusions weaken

For *Sardinella lemuru*, unreliable mitochondrial assemblies may affect fisheries genetics, population structure, and evolutionary analysis.

Existing approaches

Reference-based

Compares query sequences against known non-chimeric references.

De novo

Uses internal dataset patterns such as abundance and similarity.

Tool-based

Includes UCHIME, UCHIME2, DADA2, CATCh, and ChimPipe.

Existing tools and the study gap

	Amplicon	ASV/OTU	Read-level	Mito PCR chimera
UCHIME	Yes	Yes	Limited	No
DADA2	Yes	Yes	No	No
CATCh	Yes	Yes	No	No
ChimPipe	No	No	Yes	Different
MitoChime	No	No	Yes	Yes

Most existing tools were designed for:

- ▶ microbial amplicon or ASV/OTU-level analysis,
- ▶ reference-based or abundance-based detection,
- ▶ community-level filtering,
- ▶ biological chimeras or other sequencing contexts.

Gap addressed by this study

There is limited support for direct **read-level detection** of PCR-induced chimeras in single-species mitochondrial Illumina data.

Can machine learning detect PCR-induced chimeric reads
before mitochondrial genome assembly?

Input

Simulated clean and chimeric paired-end
Illumina reads from *S. lemuru*

Output

A modular pipeline that predicts and filters
suspected chimeric reads

Scope and limitations

Included

- ▶ *Sardinella lemuru*
- ▶ mitochondrial Illumina reads
- ▶ PCR-induced chimeras
- ▶ simulated labelled datasets

Excluded

- ▶ long-read sequencing
- ▶ NUMTs
- ▶ naturally occurring chimeras
- ▶ broad multi-species validation

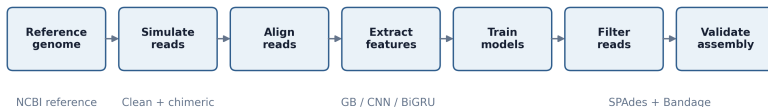
The study prioritized a focused and controlled evaluation setting.

Methodology

From simulated reads to model-based filtering

MitoChime workflow

MitoChime workflow



Overview

The workflow generated labelled reads, extracted read-level evidence, trained models, filtered suspected chimeras, and evaluated downstream assembly behavior.

Data construction

Clean reads

Simulated from the *S. lemur* mitochondrial reference genome.

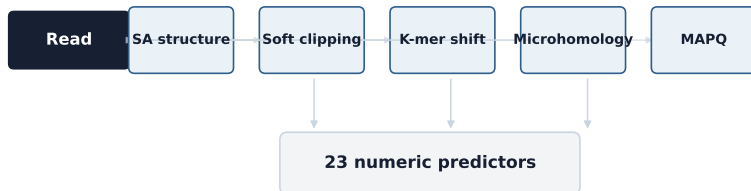
Chimeric reads

Generated by joining non-adjacent mitochondrial segments to mimic PCR template switching.

Labels: **clean** = 0 and **chimeric** = 1

Feature extraction

Evidence extracted from each read



SA structure

Soft clipping

K-mer shift

Microhomology

MAPQ

The classical machine learning models used 23 numeric read-level predictors.

Models evaluated

Gradient Boosting

Feature-based model using engineered alignment and sequence-derived predictors.

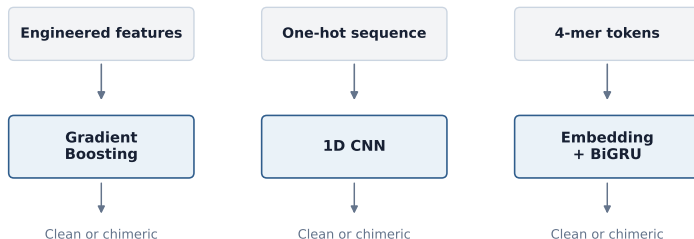
1D CNN

Sequence-based model using one-hot encoded nucleotide reads.

BiGRU

Sequence-based model using ordered k-mer tokens.

Three model representations



Evaluation strategy

Read-level evaluation

- ▶ Accuracy
- ▶ Precision
- ▶ Recall
- ▶ F1-score
- ▶ ROC-AUC

Assembly-level validation

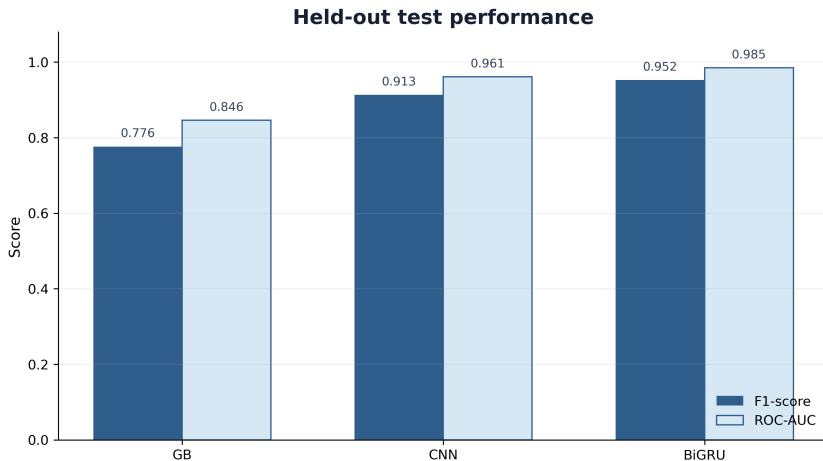
- ▶ Read retention
- ▶ Number of nodes
- ▶ Assembly graph structure
- ▶ Contiguity after filtering

The study evaluated both prediction quality and downstream utility.

Results and Discussion

Classification performance and assembly behavior

Classification performance



Model	F1-score	ROC-AUC	Runtime
Gradient Boosting	0.7765	0.8459	0.50 min
1D CNN	0.9135	0.9607	1.39 min

Model behavior

Gradient Boosting

Conservative

Higher read retention

CNN

Sensitive

More aggressive filtering

BiGRU

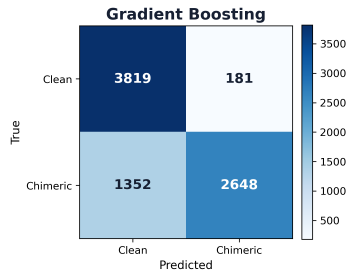
Highest F1

Best read-level performance

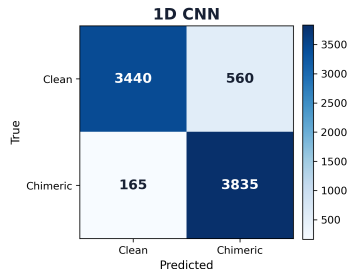
Interpretation

BiGRU achieved the best classification performance, but assembly outcomes also depended on how much read support remained after filtering.

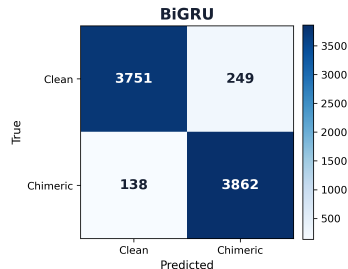
Error patterns



Gradient Boosting



CNN

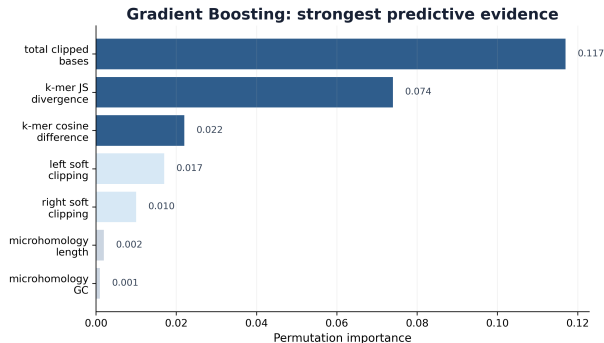


BiGRU

Main pattern

Gradient Boosting missed more chimeras but avoided over-filtering, while CNN and BiGRU detected more chimeras but removed more reads.

Feature importance



Most informative predictors:

- ▶ total clipped bases,
- ▶ k-mer Jensen–Shannon divergence,
- ▶ k-mer cosine difference,
- ▶ left soft clipping.

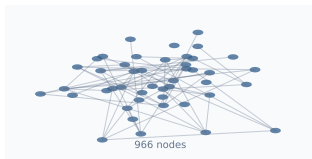
Interpretation

Chimeric reads were mainly identified through structural irregularities and sequence composition shifts.

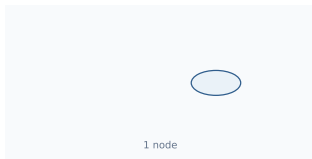
Assembly validation: high coverage

Assembly graphs: 20K reads, 50% chimeras

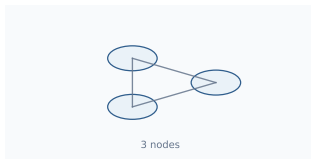
Unfiltered



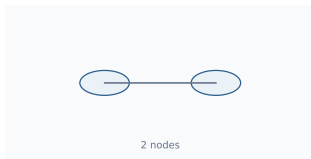
CNN



GB



BiGRU



20K reads, 50% chimeras

- ▶ Unfiltered assembly showed severe graph complexity.
- ▶ Filtering simplified the graph.
- ▶ CNN produced the simplest graph.
- ▶ GB retained more reads.

Assembly validation: low coverage

Assembly graphs: 3,200 reads, 50% chimeras

Unfiltered



2 nodes

GB



1 node

CNN



2 nodes

BiGRU



8 nodes

3,200 reads, 50% chimeras

At low read depth, aggressive filtering may remove too much read support and cause fragmentation.

Practical implication

The safest model depends on coverage and chimera proportion.

Recommendations

For practical use

- ▶ Use Gradient Boosting when read depth is low.
- ▶ Use BiGRU when read-level classification is the priority.
- ▶ Use CNN cautiously under low coverage.

For future work

- ▶ Test on real mitochondrial datasets.
- ▶ Evaluate realistic class imbalance.
- ▶ Add varied sequencing-error profiles.
- ▶ Explore hybrid feature-sequence models.

Summary

- ① PCR-induced chimeras can disrupt mitochondrial genome assembly.
- ② Existing tools do not directly address this read-level setting.
- ③ MitoChime detects suspected chimeric reads before assembly.
- ④ BiGRU achieved the strongest classification performance.
- ⑤ Model choice should consider both accuracy and read retention.

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Thank you.